Short answer: **You don't install AWX directly on Amazon Linux 2023** ❌

👉 It runs on **Kubernetes (k8s)**, so you need that first.

Simple way (recommended for learning):

1. Install Docker (for k8s)

```
sudo dnf install docker -y
sudo systemctl enable --now docker
```

2. Install k3s (lightweight Kubernetes)

```
curl -sfL https://get.k3s.io | sh -
```

3. Install kubectl

```
sudo ln -s /usr/local/bin/k3s /usr/local/bin/kubectl
```

4. Install AWX Operator

```
kubectl create namespace awx
kubectl apply -f https://raw.githubusercontent.com/ansible/awx-operator/devel/deploy/awx-operator.yaml -n awx
```

5. Deploy AWX

```
kubectl apply -f https://raw.githubusercontent.com/ansible/awx-operator/devel/deploy/awx-demo.yaml -n awx
```

Then:

👉 Access AWX via browser (get service IP/port)

Reality check:

⚠️ This is a bit heavy setup

👉 For practice, many people use **Docker Desktop / Minikube locally**

Interview one-liner:

“AWX is installed using Kubernetes via the AWX Operator, not directly on the OS.”

Yeah 👍 if you just want **quick practice**, go simple:

◇ Easiest setup (Docker-based AWX)

👉 No Kubernetes headache

1. Install Docker

```
sudo dnf install docker -y
sudo systemctl enable --now docker
```

2. Install docker-compose

```
sudo dnf install docker-compose-plugin -y
```

3. Clone AWX repo

```
git clone https://github.com/ansible/awx.git
cd awx/tools/docker-compose
```

4. Start AWX

```
docker compose up -d
```

5. Access:

👉 Open browser → `http://<EC2-IP>:8043`

⚠️ Reality:

- Not production ready
- Just for learning/testing

Even simpler (honest advice):

👉 Use **Playbooks via CLI first**, then AWX later

Interview one-liner:

"For quick setup, AWX can be run using Docker Compose, but production setups use Kubernetes."

Perfect 👍 here's your **clean, final, interview-ready AWX setup guide (Amazon Linux 2023)**

👉 simple + all fixes included

🚀 AWX Installation (Complete Guide)

☑ 1. Prerequisites

Install required tools

```
sudo dnf install -y git ansible python3-pip
```

👉 **Why:**

- git → download AWX
- ansible → run installer
- pip → install Python packages

☑ 2. Install Docker (if not done)

```
sudo dnf install -y docker
sudo systemctl enable docker
sudo systemctl start docker
```

👉 **Why:** AWX runs inside Docker containers

Fix Docker permission

```
sudo chmod 666 /var/run/docker.sock
```

👉 **Why:** lets Ansible talk to Docker

☑ 3. Install Docker Compose (binary)

```
sudo curl -L https://github.com/docker/compose/releases/download/v2.24.6/docker-compose-linux-x86_64 -o /usr/local/bin/docker-compose
sudo chmod +x /usr/local/bin/docker-compose
```

👉 **Why:** AWX uses docker-compose to create containers

☑ 4. Fix Python dependencies (VERY IMPORTANT)

Clean broken packages

```
sudo rm -rf /usr/local/lib/python3.9/site-packages/docker*
```

Install correct versions

```
sudo /usr/bin/python3 -m pip install docker==5.0.3 docker-compose==1.29.2
requests-unixsocket --ignore-installed
```

Verify

```
python3 -c "import docker, compose; print(docker.__version__)"
```

👉 Should show:

```
5.0.3
```

👉 **Why:** AWX only works with specific versions

☑ 5. Clone AWX

```
git clone -b 17.1.0 https://github.com/ansible/awx.git
cd awx/installer
```

👉 **Why:** version 17.1.0 supports Docker install

☑ 6. Configure inventory

```
nano inventory
```

Add/modify:

```
localhost ansible_connection=local ansible_python_interpreter=/usr/bin/python3

admin_user=admin
admin_password=Admin@123
```

```
host_port=80
```

👉 **Why:**

- forces correct Python
- sets login creds
- exposes port 80

☑ 7. Set Docker environment

```
export DOCKER_HOST=unix:///var/run/docker.sock
```

👉 **Why:** fixes "http+docker" error

☑ 8. Run AWX installer

```
sudo ansible-playbook -i inventory install.yml
```

👉 Takes ~5–10 mins

☑ 9. Verify installation

```
docker ps
```

👉 You should see:

- awx_web
- redis
- postgres

👉 **Why:** confirms AWX is running

☑ 10. Fix login (IMPORTANT)

If login fails → create admin manually

```
docker exec -it awx_web bash
awx-manage createsuperuser
```

👉 Enter:

- username: admin
- password: your choice

Exit:

```
exit
```

☑ 11. Access AWX UI

Open browser:

http://<your-ec2-public-ip>

⚠ Common Issues (quick fixes)

❌ docker module error

✔ install:

```
pip3 install docker
```

❌ http+docker error

✔ fix:

```
pip3 install requests-unixsocket
export DOCKER_HOST=unix:///var/run/docker.sock
```

❌ compose module error

✅ fix:

```
pip3 install docker-compose==1.29.2
```

❌ version conflict

✅ fix:

```
pip install docker==5.0.3 --force-reinstall
```

❌ login not working

✅ fix:

```
awx-manage createsuperuser
```

🔥 Final Flow (super short)

1. Install tools
2. Install Docker
3. Fix Python deps
4. Clone AWX
5. Edit inventory
6. Run playbook
7. Create user
8. Open UI

🎯 Interview One-Liner

AWX is installed using Docker containers with Ansible, requiring correct Python dependencies and is accessed via a web UI.

If you want next:

👉 I can give **how to run your first playbook in AWX (very important for interviews)** 👍